

Run 05 | OME

India

IV only

Target Opportunity-motivated entrepreneurship; binary 1 vs. 0
Sample 27,555 respondents **Coverage** 1 country; 2015–2022
Positive cases 2,167 (7.86%) **Case time** 43s
Train / val. / test 18,739 / 3,306 / 5,510 (68% / 12% / 20%, stratified)
Predictors (13): Self-efficacy, opportunity perception, fear of failure, knowing entrepreneurs, intention, media coverage, business exit, business angel, established business owner, good career choice, ease of starting a business, high status, social enterprise.
Fraser EFW predictors (7): Overall economic freedom, Economic freedom quartile, Size of government, Legal system and property rights, Sound money, Freedom to trade internationally, Regulation. Total independent variables: 20.
Controls (0): None. Country and respondent ID are excluded as predictors.
Protocol: Ten CPU models; model seed 20260925; fixed splits by sample/outcome; training-only preprocessing; F1 threshold selected on validation data. Test scores below are unweighted.

Result in brief

Highest observed ROC-AUC: H2O GBM (0.9873). Highest F1: Random Forest (0.7885). The all-negative accuracy baseline is 92.14% (full sample).

Model performance on held-out test records

Model	ROC-AUC	PR-AUC	Acc.%	Prec.%	Recall%	F1	Time
H2O GBM	0.9873	0.8488	96.15	74.07	78.52	0.7623	3s
Random Forest	0.9872	0.8510	96.26	70.98	88.68	0.7885	<1s
XGBoost	0.9871	0.8450	96.10	74.44	76.67	0.7554	<1s
CatBoost	0.9870	0.8452	96.10	70.41	86.84	0.7777	6s
Elastic Net	0.9860	0.8339	96.04	71.89	81.52	0.7641	12s
Logistic Regression	0.9859	0.8330	96.01	71.60	81.52	0.7624	<1s
Lasso	0.9858	0.8278	96.08	69.48	89.38	0.7818	7s
Decision Tree	0.9855	0.8074	95.95	71.43	80.83	0.7584	<1s
Ridge	0.9854	0.8263	95.81	70.87	79.21	0.7481	6s
Naive Bayes	0.9070	0.5732	92.89	53.55	71.36	0.6119	6s

Models ranked by ROC-AUC; small differences are descriptive, not significance tests. Time includes fit, prediction and output processing. Logistic regression uses a small stabilizing penalty; Lasso/Ridge/Elastic Net use 3-fold training CV.

Random-forest feature importance: top 8

Rank	Predictor	Raw importance	Share%
1	Entrepreneurial intention	835.77	46.68
2	Established business owner	323.71	18.08
3	Business angel	82.08	4.58
4	Knowing entrepreneurs	68.06	3.80
5	Business exit	40.52	2.26
6	Sound money	26.70	1.49
7	Regulation	26.07	1.46
8	Self-efficacy	26.04	1.45

Share is the percentage of total forest impurity importance across all 38 encoded predictors, not explained variance. Raw values are not comparable across samples. Missing indicators are derived preprocessing features.

Leading Fraser EFW predictors within the same forest

Rank	Fraser EFW predictor	Raw importance	Share%
1	Sound money	26.70	1.49
2	Regulation	26.07	1.46
3	Overall economic freedom	25.55	1.43

Fraser EFW subset ranked by raw importance; shares use total forest importance, not the Fraser EFW subtotal.

Source: Completed Objective 3 batch, 29 September 2026; 05_obj3_india_iv_OME. Scores and importance are taken directly from saved CSV outputs.